

Problem 85 (British MO 2016 Round 1 Problem 6). A positive integer is said to be *good* if it is 2 or it is of the form $3^m 5^n$ for some nonnegative integers m and n . Prove that every positive integer can be represented as the sum of different good integers.

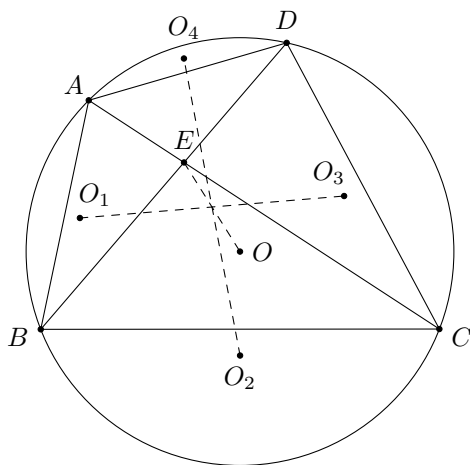
Solution. We prove the assertion by induction. The assertion holds for the positive integers $1 = 1$, $2 = 2$, $3 = 1 + 2$, $4 = 1 + 3$ and $5 = 5$. Consider the case $k \geq 6$. Let x be the largest good integer smaller than k . Then $x \geq 5$, and so we can write $x = 3^m 5^n$.

If $n \geq 1$, then $3^{m+2} 5^{n-1}$ is another good integer which is larger than x . Thus, it must be larger than or equal to k . This implies $x < k \leq 3^{m+2} 5^{n-1} < 2x$. By the inductive hypothesis, $k - x$ is a sum of different good integers. These good integers are different from x because $k - x < x$. Therefore, we can write $k = (k - x) + x$ as the sum of these good integers together with x .

If $n = 0$, then $x \geq 9$, and so $3^{m-1} 5$ is another good integer which is larger than x . As above, we have $x < k \leq 3^{m-1} 5 < 2x$, and we can also write $k = (k - x) + x$ as the sum of different good integers.

The proof is complete by induction. □

Problem 86 (Thailand MO 2017 Problem 2). Let $ABCD$ be a cyclic quadrilateral with circumcentre O . The diagonals AC and BD meet at E . Let O_1, O_2, O_3, O_4 be the circumcentres of $\triangle EAB, \triangle EBC, \triangle ECD, \triangle EDA$. Prove that $O_1 O_3$ and $O_2 O_4$ meet at the midpoint of OE .



Solution. Note that $\triangle EAB$ and $\triangle EDC$ are oppositely similar. As O_1 and O_3 are the circumcentres of these triangles, the lines EO_1 and EO_3 are isogonal with respect to $\angle AEB$. This shows $EO_3 \perp AB$ (which can also be proved by simple angle chasing). Also, we have $O_1 O \perp AB$ since $O_1 O$ is the perpendicular bisector of AB . Thus, $EO_3 \parallel O_1 O$. Similarly, $EO_1 \parallel O_3 O$. Thus, $EO_1 O O_3$ is a parallelogram, so that $O_1 O_3$ bisects OE . Similarly, $O_2 O_4$ bisects OE . This proves the result. □

Problem 87 (British MO 2017 Round 2 Problem 4). X and Y play a game as follows. Initially, X picks a 3-digit number n from 000 to 999 (possibly starting with 0) secretly. In each round, Y chooses a 3-digit number m from 000 to 999. If m and n share at least one common digit in the same position, X will tell Y this is a *good* guess. Otherwise, X will tell Y this is a *bad* guess. For example, if X has chosen 012, then 012, 020 and 314 are good guesses, while 101, 123 and 456 are bad guesses. At least how many rounds are needed so that Y can guarantee to know the value of n ?

Solution. At least 13 rounds are needed.

We first show that Y can guarantee to know the value of n in 13 rounds. In the first 10 rounds, Y chooses 000, 111, $\dots$, 999 respectively.

- If only one of these is good, say 000, then $n = 000$.

- If only two of these are good, say 000 and 111, then each digit is 0 or 1. In the next 3 rounds, Y chooses 199, 919, 991 respectively to check whether each digit is 0 or 1, hence confirming the value of n .
- If three of these are good, say 000, 111 and 222, then the digits of n are 0, 1, 2 in some order. In the next 2 rounds, Y chooses 199, 299 respectively to check whether the first digit is 0, 1 or 2. WLOG assume n starts with 0. In the last round, Y chooses 919 to check whether the second digit is 1 or 2, hence confirming the value of n .

We now show that 12 rounds are insufficient. Suppose all the first 6 guesses are bad. Then there are at least 4 possible choices for each digit. Let m be the 7th guess. If one of the digits of m has been used in one of the previous guesses (in the same position), X may tell Y this is a bad guess. Then there are at least $4 \times 3 \times 3 = 36$ possibilities for the value of n . Afterwards, note that it is possible that each guess may only reduce the number of possibilities by at most one half. Thus, after 5 guesses, there are at least $\left\lceil \frac{36}{2^5} \right\rceil = 2$ possibilities, which means Y cannot guarantee to know the value of n .

Similarly, if all digits of m have not been used, X may tell Y this is a good guess. Then there are $4^3 - 3^3 = 37$ possibilities left. Again, Y cannot guarantee to know the value of n in 5 rounds.

Therefore, 13 rounds are needed. □

Problem 88 (Turkey MO 2017 2nd Round Problem 3). Let a_{ij} be the (i, j) th entry of an $n \times n$ matrix, where n is a positive integer. Suppose $a_{ij} > 0$ and $a_{ij}a_{ji} = 1$ for any i, j (possibly $i = j$). Find the maximum value of

$$\sum_{i=1}^n \frac{1}{\sum_{j=1}^n a_{ij}}.$$

Solution. We prove that the maximum value is 1 by induction on n . The case $n = 1$ is trivial as $a_{11} = 1$.

For the inductive step, consider the case $n \geq 2$. Let $b_i = a_{in}$ so that $a_{ni} = \frac{1}{b_i}$. Note that $b_n = a_{nn} = 1$. Let $x = \sum_{i=1}^{n-1} \frac{1}{b_i}$, let $c_i = a_{i1} + a_{i2} + \cdots + a_{i,n-1}$, and let

$$S = \sum_{i=1}^n \frac{1}{\sum_{j=1}^n a_{ij}} = \sum_{i=1}^{n-1} \frac{1}{c_i + b_i} + \frac{1}{x+1}. \quad (1)$$

By the Cauchy-Schwarz inequality, we have $(c_i + b_i) \left(\frac{1}{c_i} + \frac{1}{x^2 b_i} \right) \geq \left(1 + \frac{1}{x} \right)^2$. This implies

$$\frac{1}{c_i + b_i} \leq \left(\frac{1}{c_i} + \frac{1}{x^2 b_i} \right) \left(1 + \frac{1}{x} \right)^{-2}. \quad (2)$$

Combining (1) and (2), and using the inductive hypothesis $\sum_{i=1}^{n-1} \frac{1}{c_i} \leq 1$, we obtain

$$S \leq \left(1 + \frac{1}{x} \right)^{-2} \sum_{i=1}^{n-1} \left(\frac{1}{c_i} + \frac{1}{x^2 b_i} \right) + \frac{1}{x+1} \leq \left(1 + \frac{1}{x} \right)^{-2} \left(1 + \frac{1}{x^2} \sum_{i=1}^{n-1} \frac{1}{b_i} \right) + \frac{1}{x+1} = \left(1 + \frac{1}{x} \right)^{-1} + \frac{1}{x+1} = 1.$$

Also, when $a_{ij} = 1$ for all i, j , we have $S = 1$. This shows the maximum value of S is 1. The result follows by induction. □